Sample Midterm Exam 2

Econ 5030

1 Preferences

Consider an agent with preferences over $X = [0,1] \times [0,1]$ defined as follows: For all $x, x' \in X$,

$$\begin{aligned} &x \succ x' \text{ if } x_1 > x_1', \\ &x \succ x' \text{ if } x_1 = x_1' \text{ and } x_2 > x_2', \\ &x \sim x' \text{ if } x = x'. \end{aligned}$$

1.1 Write the definition of complete preferences and clearly demonstrate that these preferences are (or are not) complete on X.

1.2 Write the definition of transitive preferences and clearly demonstrate that these preferences are (or are not) transitive on X.

1.3 Does a utility function exist that represents these preferences on X? Explain.

1.4 Now restrict the choice set to the finite set $X' = \{0,1\} \times \{0,1\}$ with the same preferences. Does a utility function exist? If so, write one down.

2 Game Theory

2.1 Timing and Strategies

Take a simple two player game with each player having two actions. Player 1's set of actions is $\{a_1, b_1\}$. Player 2's set of actions is $\{c_2, d_2\}$. The players' payoffs over the four possible action combinations are as follows:

$$\begin{aligned} u_1(a_1, c_2) &= 20 & \qquad u_2(a_1, c_2) &= 10 \\ u_1(a_1, d_2) &= 5 & \qquad u_2(a_1, d_2) &= 0 \\ u_1(b_1, c_2) &= 15 & \qquad u_2(b_1, c_2) &= 5 \\ u_1(b_1, d_2) &= 30 & \qquad u_2(b_1, d_2) &= 15 \end{aligned}$$

Suppose this is a sequential choice game in which player 1 goes first, then player 2 observes player 1's choice and makes her choice.

2.1 Write the Normal Form of the game.

2.2 Construct the Game matrix.

2.3 Find the set of strategies that remain after iterated deletion of strictly dominated strategies.

2.4 Find the set of Nash Equilibrium.

2.5 Write the Game Tree and find the set of Subgame perfect equilibrium. Which Nash equilibria are based on incredible threats?

2.2 Games with Different Preferences

There are two players and each player i has two actions $\{R_i, Q_i\}$. The money each player i receives for the different action pairs (in a game with player j) is described in the matrix below (*this is not the Game Matrix*):

		$P2$	
		R_2	Q_2
$P1$	R_1	\$8, \$8	\$2, \$12
	Q_1	\$12, \$2	\$0, \$0

The game is sequential. Player 1 picks her action first and then player 2 observes the action of player 1 and player 2 picks his action. We will consider 3 cases of this game.

Case 1: Both players only care about their own money with more money being strictly better than less.

2.6 Create a normal form for this game. Write the game matrix and find the set of Nash Equilibrium. Draw the game tree and find the set of Subgame perfect equilibrium.

Case 2: Each player only cares about the sum of the two players' money and strictly prefers more total money to less.

2.7 Create a normal form for this game. Write the game matrix and find the set of Nash Equilibrium. Draw the game tree and find the set of Subgame perfect equilibrium.

Case 3: Player 1 only cares about her own money. Player 1 does not know Player 2's type. With probability $\mu \in (0, 1)$, Player 2 is *spiteful*: his payoff is his own money minus Player 1's money. With probability $1 - \mu$, he is selfish (cares only about his own money). Player 2 knows his own type.

2.8 Create a game tree for this game and use a form of backward induction as your solution concept. For which values of μ does Player 1 choose Q_1? Explain clearly the difficulties you faced in finding a way to analyze the game.

3 Conceptual Questions

3.1 What is expected utility? What extra assumption (beyond complete, transitive and continuous preferences over lotteries) is required for an expected utility representation? Give an example of behavior that violates it.

3.2 Define a best response and a (pure strategy) Nash Equilibrium. What does Nash Equilibrium assume that rationalizability does not?

3.3 In a sequential game, what is a strategy? Why does the second mover have more strategies in the sequential version of a game than in the simultaneous version, even though she has the same actions?

3.4 Define consumer surplus, producer surplus and total surplus in a market game. Why is total surplus a natural starting point for evaluating welfare, and why is it not a complete answer?

4 Cournot Oligopoly

Consider a Cournot oligopoly with 3 firms. Firms simultaneously choose quantities $q_i \geq 0$ of a homogeneous good. Each firm has cost $c(q_i) = 10q_i$. Market inverse demand is $P(Q) = 100 - Q$, where $Q = q_1 + q_2 + q_3$ and $Q_{-i} = Q - q_i$.

4.1 Write the Normal Form of this game.

4.2 Find the best response function of firm i, $b_i(Q_{-i})$, and graph it on the Q_{-i}–q_i plane.

4.3 Find the Cournot Nash equilibrium quantity of each firm q^c, total output Q^c and price p^c.

4.4 Find each firm's profit, CS, PS, and TS.

4.5 Find the efficient outcome. Define and compute the Dead Weight Loss. Draw the market on the p–Q plane showing CS, PS and DWL.

4.6 Suppose the three firms form a cartel that maximizes joint profit and splits output equally. Find the cartel output and each firm's profit. Is this a Nash equilibrium? Explain.

4.7 Generalize to n firms with the same demand and costs. Find $q^c(n)$, $p^c(n)$ and $DWL(n)$. What happens as n grows? What does this say about the effect of competition on efficiency?